

Chirag Gupta

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AI engineer building multi-agent systems and the infrastructure that makes them measurable: cost accounting, evaluation, and runtime security for LLM agents.

EXPERIENCE

AI Engineer, GoManzanas (US Startup)

May 2025 - Present

Remote

AEGIS, Multi-Agent Pipeline: Detected pirated CAD software usage by analyzing telemetry and signal patterns.

- Staged multi-agent pipeline over multi-backend LLM orchestration (fallback routing, stage-specific prompting): 89% precision against a 94% manual baseline at \$0.05/case, down from \$1.06

Ace1t, Real-Time Voice AI: Built voice-AI backend for a real-time accent training platform.

- Built the streaming push-to-talk pipeline end to end: multiple TTS models behind Whisper and FastConformer ASR, model-driven speech control, silence handling

Booster, AI Collaboration Platform: Built multi-agent infrastructure for an ideation platform (FastAPI + React + PostgreSQL).

- Built multi-agent group chat with turn-based orchestration and real-time collaboration across agent personas
- Designed containerized code execution environment and agent session lifecycle with idle cleanup and queue-based concurrency
- Built custom MCP server, skills system with model tiering, and agent benchmarking framework

GSoC '25 Contributor | '26 Mentor, Jenkins

May 2025 - Present

- Built modular multi-agent diagnostic system (log-parser, classifier, recommender) for Jenkins build failures, across OpenAI, Gemini, and Claude backends
- Added vector-store RAG layer, benchmarked on 20 curated questions: 95% context relevance, 3.75/5.0 overall score

AI Engineer, Reasonify Technology Pvt Ltd

- Built multi-agent LLM pipelines for adaptive learning content and automated study-plan generation
- Built automated Manim video generation pipeline (CLI + prompt standardization + component library), cutting production time from 5h to 20min
- Implemented hybrid semantic filtering for TTS using regex rules and a fine-tuned Gemma 270M model

Open Source

- Merged upstream fixes: keephq/keep, datalayer/jupyter-mcp-server (stale-tool fallback and concurrent-miss coalescing), topoteretes/cognee, HKUDS/DeepTutor
- Rago (OSL Incubator): Added 5 API backends (Cohere, Fireworks, Together, Groq, HuggingFace) for embeddings and generation. 7 merged PRs.
- Keras: np.diff and np.log2 via the OpenVINO backend. Agno: Exa Answer method, deprecated-model migration across 7+ providers, Crawl4ai URL expansion.

AI & Data Science Intern

May 2024 - Jan 2025

Mindcraft Labs | EvoAstra Ventures

- Mindcraft Labs:** automated 19+ workflows (RETELL, Voiceflow, Make). **EvoAstra:** fraud-detection model on 6.3M+ records at 94% recall

PROJECTS

Unplug: Open-source runtime security for AI agents. Trained and shipped unplug-tiny, a DeBERTa prompt-injection span detector (94.4% recall at 0.5% FPR) that redacts the malicious span instead of dropping the whole document. 3-stage pipeline with taint tracking across tool calls; Python SDK on PyPI plus a hosted API. unplug-ai.org · github.com/UnplugAI/Unplug

Hive: Local-first agent OS. Persistent agents from YAML, multi-model routing (Claude, Codex, LM Studio), agent rooms for collaboration.

github.com/chiruu12/Hive

Marshal: Runs a fleet of headless coding agents (Claude Code, Codex, Cursor, OpenCode) in parallel, each in an isolated git worktree, with a per-provider cost ledger. CLI, MCP server, and Claude Code plugin; on PyPI as marshal-agents. Same task, four backends, priced from the ledger: the cheapest correct run cost 1/115th of the most expensive. github.com/chiruu12/marshal

EDUCATION

M.Sc. (Hons.) Mathematics & B.E. (Hons.) Electronics & Instrumentation

2021 - 2026

BITS Pilani, Goa Campus, Goa, India

SKILLS

Languages: Python, C/C++, SQL

ML/AI: PyTorch, Hugging Face Transformers, ONNX Runtime, vLLM, LangChain, MLflow

Infrastructure: Docker, Git, FastAPI, Pydantic, PostgreSQL, MCP, gRPC

ACHIEVEMENTS

- Top 20, Meta Hacker Cup Open AI practice round (13,000+ entrants)
- Winner, Smart India Hackathon 2025
- 4th place, ContextCon, YC Startup School hackathon